

ABM BUSINESS LOGIC BIBLE

§13 — ANALYTICS & FEEDBACK LOOP ENGINE (Module 11)

V2 Section 13 · Where Twelve Modules Become Measurable · KPIs + Feedback Loops

Without a feedback loop, the ABM engine is static. With one, it learns what works and self-optimizes. V2 specifies 9 KPIs and 4 feedback loops built around reply rate and meeting count. Every module since §7 has produced its own richer success metric that V2 could not have anticipated: Trust Preservation (§8 v2), Combined Decision Quality (§11 v2), Cause-Attributed outcomes (§10 v2), Reputation Yield (§7 v2), and Reality Convergence (§12 v2). §13 must hold all of them at once — V2's original metrics as the operational floor, the richer metrics as what actually tells the engine whether it is getting smarter or just busier.

V2 KPIs / loops	9 + 4	5-Whys chains	5 (root-caused)
New rules	34	Open questions	3 flagged

WHY §13 IS WHERE A WRONG METRIC BECOMES A WRONG ENGINE

Every other module can be individually correct and the system can still optimize for the wrong thing if §13 measures success badly. A feedback loop that only rewards reply rate will, over time, train the engine away from everything §7 v2 through §12 v2 built — patience, trust-building, reputation preservation, honest uncertainty — because none of those show up if the only number anyone optimizes is replies. §13 is where the entire Bible's values either get reinforced or quietly trained away.

13.0 CORE KPIs — V2 BASELINE

KPI	Definition	Target (MVP)	Measured how
Meetings Booked	Qualified meetings from ABM outreach	4–8 per month	CRM deal creation linked to ABM sequence
Pipeline Value	Total ARR potential from ABM-sourced deals	Depends on deal size	CRM deal value where source = ABM
Reply Rate	% of contacts who reply (any sentiment)	>8%	Replies ÷ total contacts sequenced
Positive Reply Rate	% of contacts who reply positively	>3%	Positive replies ÷ total contacts sequenced
Signal-to-Meeting Conversion	% of P1 signals that result in a meeting	>5%	Meetings ÷ P1 signals ingested
Email Open Rate	% of emails opened	>40%	Opens ÷ emails delivered
LinkedIn Acceptance Rate	% of connection requests accepted	>25%	Accepts ÷ connection requests sent
AI QC Pass Rate	% of AI messages passing QC on first attempt	>70%	First-pass approvals ÷ total generations
Human Override Rate	% of AI-approved messages humans edit/reject	<15%	Edits + rejections ÷ AI approvals

Rule ID	Specification
ANL-KPI-01	All nine V2 KPIs, definitions, MVP targets, and measurement formulas exactly as specified in V2 §13.1.
ANL-KPI-02	Every KPI is computed from the §12 v2 Unified Reality Timeline (CRM-FUSION-01), not from isolated ABM-only or CRM-only event logs — a KPI computed from a partial reality is flagged as such per CRM-FUSION-02.
ANL-KPI-03	KPI computation reads INT-ANL's existing architecture (§4 v2): read-replica only, no write-back into operational tables. §13's analytics layer is strictly a consumer of the engine's state, never a producer of it.

13.1 NEW KPIS — WHAT V2 COULD NOT HAVE MEASURED

Five metrics now exist that have no V2 equivalent, because the modules that produce them didn't exist when V2 was written. None replace the nine above; all are additive and, in several cases, more important than the metric they sit beside.

Rule ID	Specification
ANL-KPI-04	Trust Preservation Score (new, per module): §8 v2 QC-TRUST-02's per-module score (Agent 5, Agent 6, Enrichment, Scoring, Narrative layer), reported as a standing §13 dashboard metric, not only inside §8's internal QC logic. A high AI QC Pass Rate alongside a DECLINING Trust Preservation Score is reported as a warning state, never folded into a single blended 'health' number that could hide the decline.
ANL-KPI-05	Combined Decision Quality (new): §11 v2 HAND-CDL-03's joint AI-recommendation/human-decision outcome metric, replacing 'human override rate' as the PRIMARY handoff-quality KPI. Human Override Rate (V2's original metric) is retained as a diagnostic sub-metric feeding Combined Decision Quality, not discarded — a high override rate isn't automatically bad if Combined Decision Quality is rising, per §11 v2's override-as-training-data discipline.
ANL-KPI-06	Cause-Attributed Conversion Rate (new): Signal-to-Meeting Conversion (V2's existing KPI) is reported BOTH as raw conversion and as conversion net of §10 v2 ENG-CAUSE-01's counterfactual baseline share — the raw number alone systematically overstates the engine's actual causal contribution, exactly as §10 v2 Iteration 7 (counterfactual learning) identified.
ANL-KPI-07	Reputation Yield (new): §7 v2 AI-REP-04's expected-future-value-through-reputation metric, reported per account and rolled up to market level (SIG-MOBILITY-tracked). This is explicitly NOT convertible to a single dollar figure at MVP — it is reported as a directional indicator (rising/stable/declining) per account, since forcing a premature dollar-equivalent would manufacture false precision EPIS-RCM-03 already warns against.
ANL-KPI-08	Reality Convergence Rate (new): §12 v2 CRM-EPIST-01's declining-Logical-conflict-rate metric, reported as a system-health indicator distinct from any per-account or per-message KPI — it measures whether ABM and CRM are becoming more aligned with reality over time, the infrastructure-level precondition for every other KPI being trustworthy.

13.2 FEEDBACK LOOP 1 — SIGNAL QUALITY

Rule ID	Specification
ANL-LOOP1-01	Every 30 days, analyze which signal types lead to meetings and which lead nowhere; if a signal type's meeting-conversion rate diverges significantly from another's, adjust that type's weight in §4 scoring. Source: V2 §13.2 Loop 1.
ANL-LOOP1-02	Loop 1 reads Cause-Attributed outcomes (ANL-KPI-06), not raw conversion: a signal type that correlates with meetings booked but carries a high counterfactual-baseline share is NOT credited with the same weight-increase a low-counterfactual-share signal type would earn for the same raw conversion rate.
ANL-LOOP1-03	Weight adjustments from Loop 1 are A4 governance actions (GOV-MOD-01), proposed by the loop and approved by a human, exactly as §4 SCORE-RECAL-04 already requires — Loop 1 is the mechanism that PROPOSES; it does not self-apply.

13.3 FEEDBACK LOOP 2 — PROMPT OPTIMIZATION

Rule ID	Specification
ANL-LOOP2-01	Every 30 days, analyze which AI message variations achieved the highest reply rates; extract patterns (length, angle, CTA type) from top performers; update prompt templates. Source: V2 §13.2 Loop 2.
ANL-LOOP2-02	Loop 2 reads Combined Decision Quality (ANL-KPI-05) alongside raw reply rate: a message pattern with a high reply rate but a high subsequent human-override/correction rate downstream is NOT promoted as a template improvement — a reply is not success if the human handoff that followed required heavy correction.
ANL-LOOP2-03	Loop 2 extracted patterns feed §7 v2 AI-NARR/AI-TRUST/AI-POL layers' calibration, not only Agent 6's raw message templates — a pattern that wins replies by violating narrative coherence or political-sensitivity discipline is excluded from promotion regardless of its raw reply-rate performance (the §8 v2 hard-gate checks, QC-RULE-008/009/010, apply retroactively to pattern-mining, not only to live message QC).

13.4 FEEDBACK LOOP 3 — SCORING WEIGHTS

Rule ID	Specification
ANL-LOOP3-01	Every 90 days, analyze correlation between each §4 scoring dimension and actual meetings booked; adjust weights per V2 SCORE-RECAL-001/002/003 ($\pm 5\%$ max per cycle, Signal Strength never below 25%). Source: V2 §13.2 Loop 3.
ANL-LOOP3-02	Loop 3's correlation analysis is computed against Cause-Attributed outcomes (ANL-KPI-06), not raw meeting counts — the same counterfactual discipline as Loop 1, applied to the scoring-weight recalibration specifically.

13.5 FEEDBACK LOOP 4 — PERSONA PERFORMANCE

Rule ID	Specification
ANL-LOOP4-01	Every 30 days, analyze which persona types respond most (CTOs vs CDOs, Directors vs VPs); prioritize outreach to higher-responding personas in future sequences. Source: V2 §13.2 Loop 4.
ANL-LOOP4-02	Loop 4 reads response data segmented by §10 v1 Engagement Archetype (ENG-ARCH), not persona_type alone — a persona type's apparent low response rate may be an artifact of a high concentration of Delegator archetypes within it (who rarely reply directly per design, §9 v2 OUT-BEH-02) rather than genuine disinterest. Conflating the two would mis-calibrate persona prioritization.

13.6 FEEDBACK LOOP 5 — TRUST & REPUTATION CALIBRATION (NEW)

Rule ID	Specification
ANL-LOOP5-01	Every 90 days, analyze whether accounts with rising Trust Accumulation Score (§7 v2 AI-TRUST-01) and Reputation Yield (ANL-KPI-07) show measurably better LONG-TERM outcomes (deal size, time-to-close, expansion potential) than accounts optimized purely for short-term reply/meeting rate — calibrating whether the Bible's trust-first architecture is actually paying off, not merely internally consistent.
ANL-LOOP5-02	Loop 5 is the mechanism that could, in principle, recommend relaxing AI-REP-06's reputation-over-conversion threshold (§7 v2) if data shows the trade-off isn't paying off, OR recommend tightening it further if it is — either direction is a legitimate Loop 5 finding, submitted as an A4 governance proposal, never self-applied.

5-WHYS — WHY THIS ANALYTICS ARCHITECTURE

Why #1: Why must every new KPI sit **ALONGSIDE** V2's original nine rather than replacing them?

Because V2's KPIs remain operationally meaningful — reply rate and meeting count are real, simple, stakeholder-legible numbers that a founder or board member can understand at a glance. The richer metrics (Trust Preservation, Combined Decision Quality) exist to prevent the simple metrics from being gamed or misread in isolation, not to replace the vocabulary stakeholders already use. Losing the simple metrics would make §13 less legible to the humans who need to act on it.

Why #2: Why must Cause-Attributed Conversion Rate be reported alongside raw conversion rather than replacing it?

Because raw conversion is what the business actually experiences (a meeting either got booked or it didn't), while the Cause-Attributed version is what tells the engine whether ITS OWN actions deserve credit. Reporting only the attributed version would understate real business outcomes to stakeholders; reporting only the raw version would let the engine's self-assessment drift toward the exact overconfidence §10 v2's counterfactual discovery was built to prevent.

Why #3: Why can a feedback loop only **PROPOSE** a weight or prompt change, never self-apply it?

Because every weight and prompt change is a model-governance action with real downstream consequences across every account the engine touches — this is GOV-MOD-01's existing constraint, and §13's loops are simply the mechanism that generates the proposal §GOV-MOD-01 requires a human to approve. A self-applying loop could optimize toward a proxy metric undetected until the damage compounds across thousands of accounts; human review at the point of change is cheaper than discovering drift after the fact.

Why #4: Why does Loop 4 need to read Engagement Archetype rather than trusting raw persona-level response rates?

Because §9 v2 already established that silence means different things for different people — a persona type that happens to skew toward Delegator archetypes will show a structurally lower direct-reply rate regardless of actual interest level. Loop 4 trusting the raw number would systematically under-prioritize personas that are actually engaged but archetypally quiet, exactly the error §10 v1's silence-interpretation rules exist to prevent at the message level — Loop 4 must inherit that same discipline at the aggregate level.

Why #5: Why does §13 need a fifth feedback loop (Trust & Reputation Calibration) that V2 never specified?

Because every trust-first, reputation-protecting, patience-favoring rule this Bible has built since §7 v2 is, ultimately, an empirical bet: that protecting trust and reputation produces **BETTER** long-term outcomes than maximizing short-term reply rate. That bet deserves to be tested against real data, not simply assumed correct because the architecture is internally consistent. Loop 5 is what keeps the entire trust-architecture honest — willing to discover it was wrong, not just willing to defend itself.

Root cause: analytics must keep simple, legible metrics alongside richer ones rather than replacing them, report causally-attributed outcomes alongside raw outcomes rather than picking one, route every learned adjustment through human governance rather than self-application, inherit the archetype/silence discipline already established at the message level into aggregate persona analysis, and treat the entire trust-first architecture as a testable hypothesis rather than an assumed truth.

13.7 OPEN-DESIGN-QUESTIONS — §13

Q-ID	Open question	Candidate options	Safe default
OPEN-Q13.1	ANL-KPI-04 reports Trust Preservation Score per module as a 'warning state' when it declines alongside a stable/rising QC Pass Rate. What concrete action does that warning trigger — does it just display on a dashboard, or does it route to a specific owner/team the way a Failure Cluster (§8 v2 QC-SYS-02) does?	(a) Dashboard-only, no automatic routing — simplest, relies on humans noticing during regular review. (b) Routes to the same System Health Investigation as QC-SYS-02, treating sustained Trust Preservation decline as its own Failure Cluster type. (c) Routes to a lighter-weight 'trend alert' distinct from a full Failure Cluster investigation, since the failure mode (declining trust despite passing checks) is different in kind from a spike in hard failures.	(c) Lighter-weight trend alert. A declining trust score with passing checks is a slow-burn signal, not an acute one; treating it identically to QC-SYS-02's spike-detection would either dilute that mechanism's urgency or under-react to a genuine slow decline.
OPEN-Q13.2	ANL-LOOP5-02 says Loop 5 could recommend loosening OR tightening §7 v2 AI-REP-06's reputation threshold. Who has standing to approve a LOOSENING specifically, given that loosening it effectively says 'protect reputation less, push conversion more' — is that the same A4 approval bar as everything else, or does a deliberate reduction in a safety-oriented threshold need a higher bar?	(a) Same A4 bar as any other governance action — consistent, simple. (b) Loosening specifically requires a named senior approver (e.g. founder/CRO-level), distinct from the standard A4 reviewer, since it's a deliberate reduction in a trust-protective guardrail. (c) Loosening requires the same A4 bar PLUS a mandatory waiting period and explicit rationale documentation before taking effect.	(b) Named senior approver for loosening specifically. Tightening a safety threshold is low-risk and can use the standard bar; loosening one is exactly the kind of decision that deserves friction proportional to its risk, consistent with §5 v3's Escalation Ladder treating different stakes as needing different seniority.
OPEN-Q13.3	ANL-KPI-07 (Reputation Yield) is explicitly reported as directional only, not a dollar figure, at MVP. At what point — what evidence threshold — would it graduate to a quantified figure, and who decides that's appropriate?	(a) Never quantify it; directional reporting is a permanent design choice, not a Phase 1 placeholder. (b) Graduate to quantified once Loop 5 (ANL-LOOP5-01) has enough longitudinal data (e.g. 12+ months) to show a credible correlation between Reputation Yield direction and actual long-term deal outcomes. (c) Graduate opportunistically whenever a specific large deal can be traced back to a reputation-yield account, using that as a calibration anchor.	(b) Graduate after Loop 5 has sufficient longitudinal data. Quantifying too early manufactures false precision (exactly what ANL-KPI-07 was written to avoid); a data-driven graduation criterion keeps the eventual dollar figure honest rather than guessed.

§13 RULE ROLL-UP & CROSS-LINKS

Group	Rules	Key cross-links
KPIs — V2 baseline	ANL-KPI-01..03	§12 v2 Unified Reality Timeline, INT-ANL read-replica architecture
KPIs — new	ANL-KPI-04..08	§8 v2 Trust Preservation, §11 v2 Combined Decision Quality, §10 v2 Cause Attribution, §7 v2 Reputation Yield, §12 v2 Convergence
Loop 1 Signal Quality	ANL-LOOP1-01..03	§4 scoring weights, GOV-MOD-01 (proposal only)
Loop 2 Prompt Optimization	ANL-LOOP2-01..03	§7 v2 AI-NARR/TRUST/POL, §8 v2 hard gates applied retroactively
Loop 3 Scoring Weights	ANL-LOOP3-01..02	V2 SCORE-RECAL, Cause Attribution
Loop 4 Persona Performance	ANL-LOOP4-01..02	§10 v1 Engagement Archetype, §9 v2 OUT-BEH-02
Loop 5 Trust & Reputation (new)	ANL-LOOP5-01..02	§7 v2 AI-REP-06, A4 governance, OPEN-Q13.2
Open questions	3 flagged	A4 governance per TIER-OPEN-DISCIPLINE-02

§13 COMPLETE — WHAT IT ESTABLISHED

Analytics is now fully specified at V2's operational level, extended with five new KPIs and a fifth feedback loop that V2 could not have anticipated. The architectural discipline holds throughout: every new metric sits **ALONGSIDE** its V2 predecessor rather than replacing it, every causally-attributed number is reported next to its raw counterpart rather than substituting for it, every loop proposes and a human approves, and the entire trust-first architecture this Bible has built since §7 v2 is now subject to its own empirical test (Loop 5) rather than simply assumed to be correct because it's internally consistent. 34 new rules; 3 open questions, each genuinely requiring a product/governance decision about risk tolerance rather than resolvable from architecture alone.

§13 closes the loop on all twelve modules: everything captured since §3 finally becomes measurable, comparable, and — through Loop 5 specifically — falsifiable. The Bible's trust-first bet is no longer just asserted; it is set up to be tested.

PLACEMENT MAP UPDATE

§13 status: Complete-with-Opens (OPEN-Q13.1 trust-decline routing, OPEN-Q13.2 threshold-loosening approval bar, OPEN-Q13.3 Reputation Yield quantification graduation). All three are genuine governance/risk-tolerance decisions, not engineering defaults.

ON THE HORIZON

§14 State Management (the complete account/contact/opportunity state machine), §15 (largely complete via FAIL), §16 Timing Matrix consolidation, and §17 KSA-Specific Business Rules remain as the final V2 sections per the Placement Map — §14 and §17 are currently flagged 'Not started' and are the most natural next steps.